

# DOCUMENT: SOLUTION ARCHITECTURE

| Phase         | Project Design Phase                               |
|---------------|----------------------------------------------------|
| Document      | Solution Architecture                              |
| Project Name  | <b>PetPaws – Pet Foods &amp; Accessories Store</b> |
| Team ID       | [TEAM-ID]                                          |
| Date          | 14 June 2026                                       |
| Maximum Marks | 4 Marks                                            |

## Solution Architecture — PetPaws

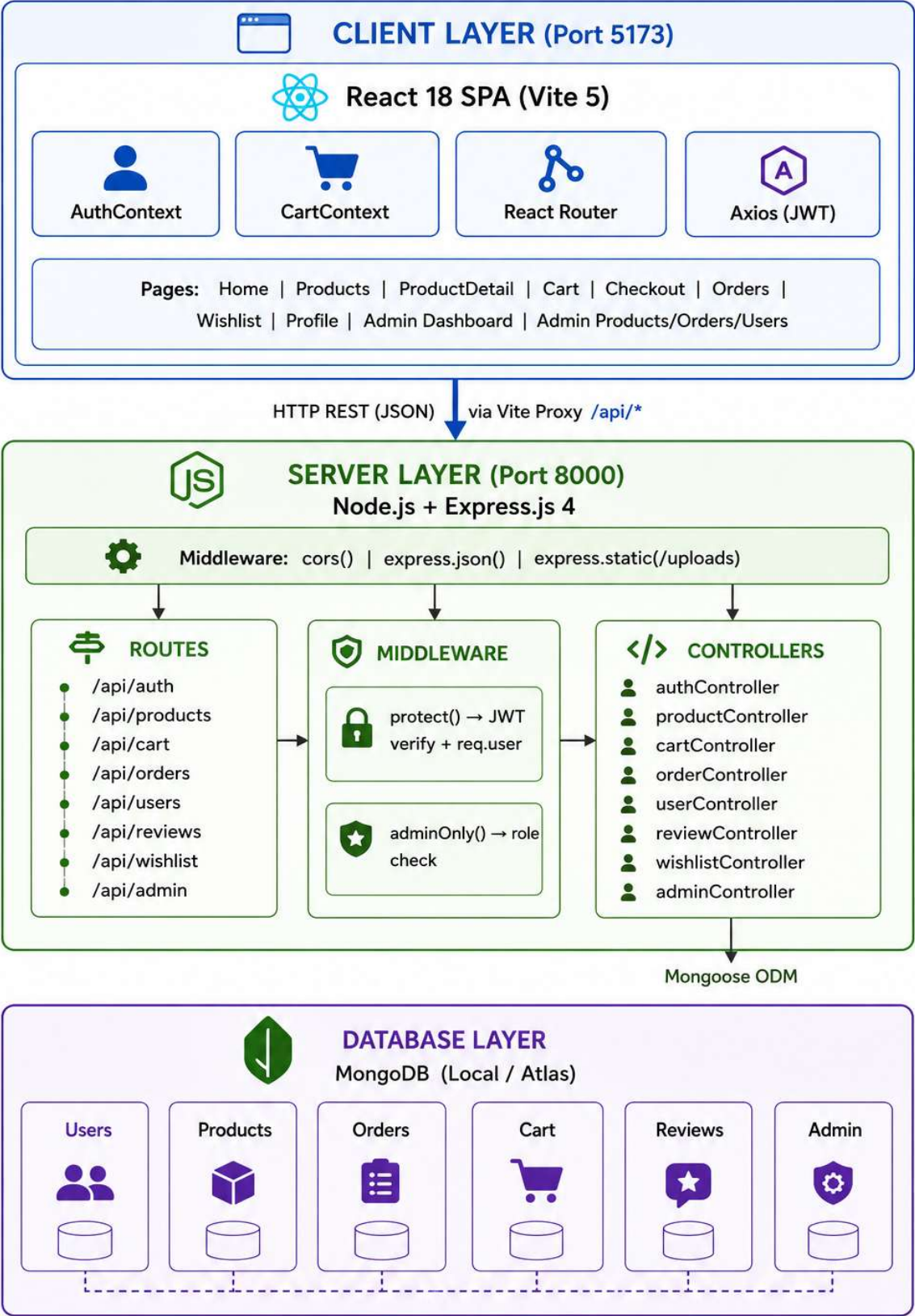
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The PetPaws architecture follows a classic three-tier MERN (MongoDB, Express.js, React.js, Node.js) pattern with clear separation between the Presentation Layer, the API/Business Logic Layer, and the Data Access Layer. The architecture is designed for horizontal scalability, stateless authentication, and independent deployment of each tier.

Architecture Diagram (Text Representation)



SOLUTION ARCHITECTURE — PetPaws



# Architecture Layers

| Layer                      | Components & Technologies                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Presentation Layer         | React 18 SPA (Vite 5)   AuthContext   CartContext   React Router v6   Axios   react-hot-toast   react-icons                       |
| API / Business Logic Layer | Node.js (v18+)   Express.js 4   JWT Middleware (protect, adminOnly)   8 Controller modules   CORS   express.json   express.static |
| Data Access Layer          | Mongoose ODM v8   MongoDB (local / Atlas)   6 Mongoose models   Text index   Aggregation pipeline                                 |
| Security Layer             | bcryptjs (salt:10)   JWT (jsonwebtoken)   Role-Based Access Control   Password select:false   Soft-delete   CORS whitelist        |
| Deployment Target          | Frontend: Vercel   Backend: Railway / Render   Database: MongoDB Atlas   Dev proxy: Vite /api/* → localhost:8000                  |

Mermaid Architecture Diagram Code

